Softwarica
COLLEGE OF IT & E-COMMERCE

in collaboration with

Coventry University

ST7072CEM MACHINE LEARNING · COURSEWORK

Analysis and Prediction of Monsoon Hazards in Nepal Using Machine Learning Approaches

Will the monsoon hurt someone in this district in the next three days?

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BACKGROUND

Every monsoon costs lives and homes

2,561
deaths from 2017 to 2025

8,586
houses destroyed

4,161
people injured

AND IN 2026

The Bhotekoshi flood in Rasuwa (26 August) killed **more than 1,300** people, with thousands missing: more than 2021 to 2025 combined.

Deaths per year from floods, landslides, heavy rain and lightning

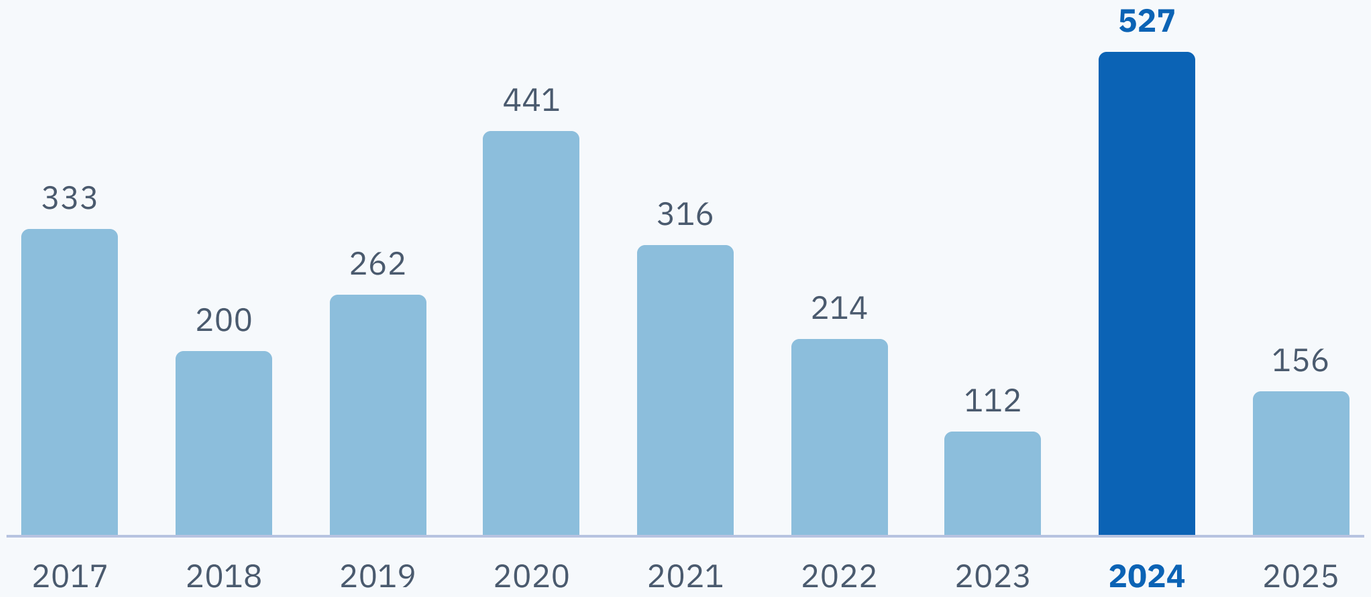


Chart: BIPAD disaster record, 2017 to 2025. 2026 flood: NDRRMA figures, 6 September 2026 (1,341 dead, 5,083 missing)

PROBLEM STATEMENT

Acting early needs a forecast of harm

THE POLICY

As per Nepal's **National Anticipatory Action Framework 2026**, relief is released **before** a hazard strikes, once pre-agreed forecast triggers are met.

WHAT IT NEEDS

Forecasts and triggers are one of its three building blocks. A trigger needs to know **which district** will be hurt, and **when**.

WHAT EXISTS

Machine learning in Nepal has mostly produced susceptibility maps: **where** landslides or floods can happen, not when.

The gap: existing work does not forecast, district by district and day by day, whether the monsoon will cause a death, injury, missing person or destroyed house in Nepal.

Source: NDRRMA, National Anticipatory Action Framework 2026, Government of Nepal

AIM AND OBJECTIVES

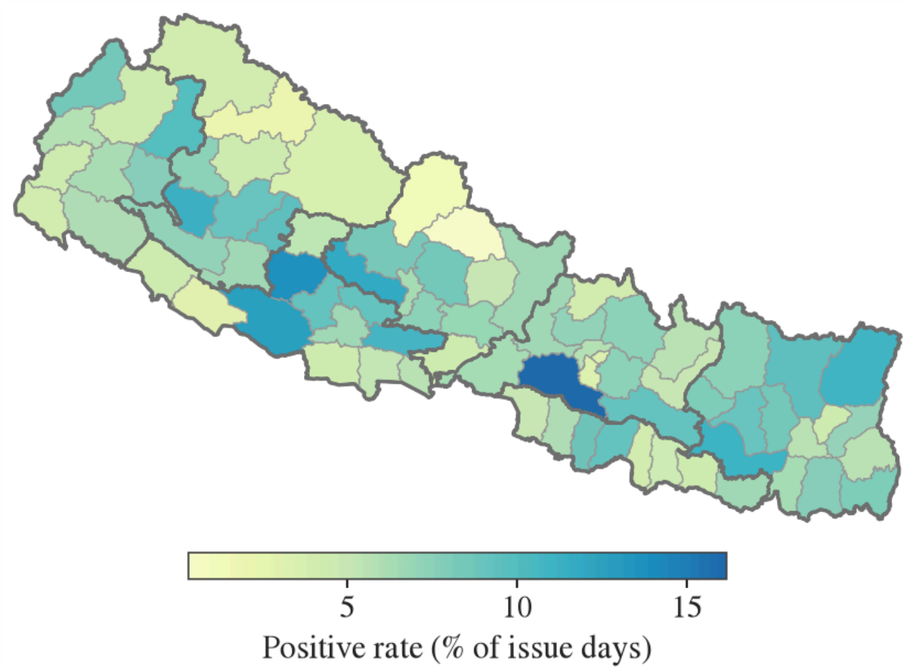
Can machine learning tell when and where monsoon harm will happen?

1. **Cluster** the 77 districts into risk regions with similar harm history and rain climate
2. **Classify** whether a district will suffer a damaging incident in the next three days
3. **Test** whether the regions improve the forecast, and explain what drives the risk

> If the forecast works, it can serve as a trigger for early relief.

DATASET

Built from open records and rain



Share of days followed by harm, by district

16,790

flood, landslide, heavy-rain and lightning records from BIPAD

146,223

district-days, April to October 2017 to 2025, with ERA5 rain

6.8%

followed by harm, so the classes are very unbalanced

> Harm rises with rain, from 3.3% of days in the driest tenth to 14.7% in the wettest. Features only use what was known on the day.

METHODS AND EXPERIMENTAL SETUP

Two stages, tested on unseen seasons

STAGE 1

Group districts

K-means, Ward and a Gaussian mixture find 6 risk regions

STAGE 2

Forecast harm

Logistic regression, random forest, gradient boosting, two SVMs

TEST

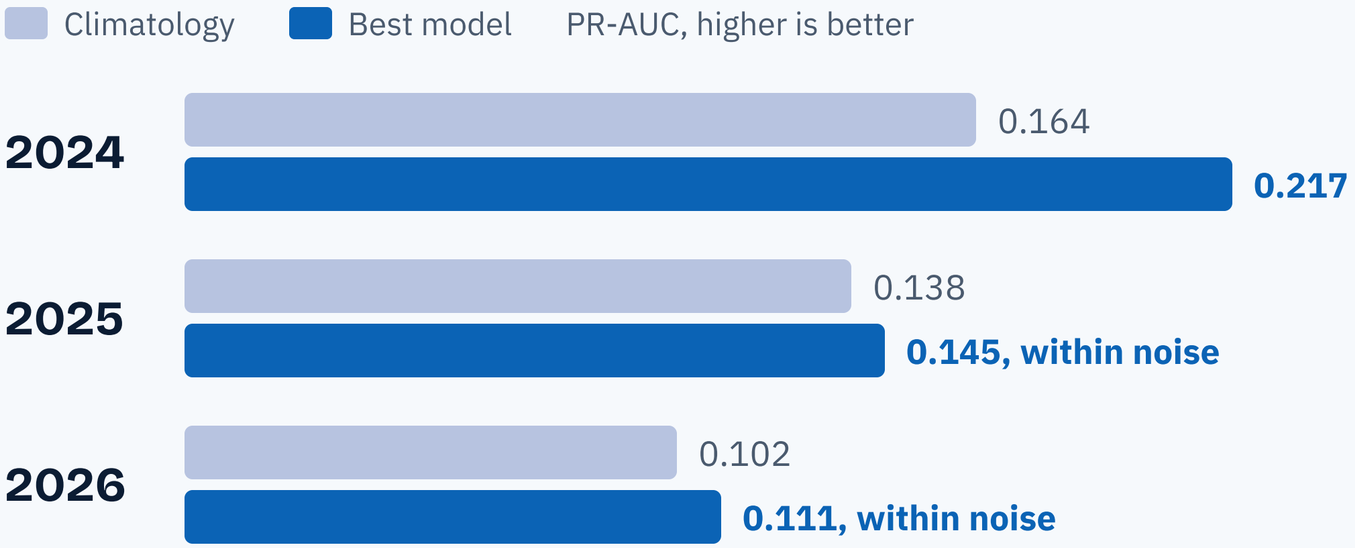
Beat the usual

Against climatology: each district’s normal rate for that month



RESULTS

A real gain, but only in one season



+0.026

mean PR-AUC gain over climatology in 2024 and 2025

Logistic regression, gradient boosting and linear SVM

DISCUSSION AND CONCLUSIONS

What each objective found

1 · Risk regions

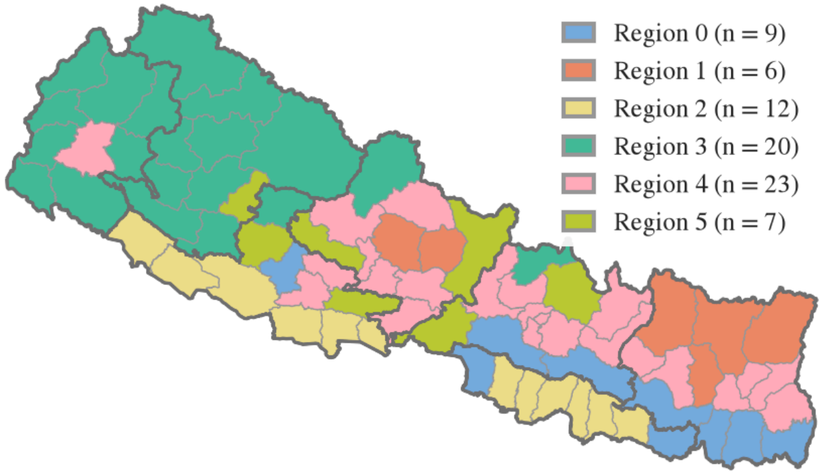
Six regions, moderately stable and easy to read on a map

2 · Forecast

Beats climatology by about 0.026 PR-AUC, but only in 2024

3 · Do regions help?

No. Rain matters more: a perfect rain forecast lifts PR-AUC to about 0.21



The six risk regions (K-means)

A harm forecast for Nepal is possible, but its value depends on **better rain forecasts** and on testing it season by season.

CODE WALKTHROUGH

Now, the code

Python and Jupyter notebooks, from raw records to the results in the paper.

GITHUB REPOSITORY
bhimrazy/st7072cem-ml-assignment

01 **Set up and check**
Install the environment, run the leakage checks

02 **Build the data**
District-day panel, exploration, risk regions

03 **Train and evaluate**
Five classifiers, calibration, explanations

04 **See the results**
The season-by-season table and main figure

LIMITATIONS

BIPAD records reported harm, so reporting bias shapes the label

Lightning is the hardest hazard to forecast from daily rain

A live system would need rain forecasts, not observed rain

Thank you

github.com/bhimrazy/st7072cem-ml-assignment

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